ALLONZO PENSABENE

BUILDING AUTOMATION & ENERGY ENGINEER

Apprentice engineer, ESTP Paris, class of 2027

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5 years of apprenticeship, always school and industry at once. Solar PV first, BACS/BMS now, and the software I build around it.

OPEN TO ENGINEERING ROLES ABROAD
Mobility from March 2027 · full time from October 2027

EXPERIENCE

APPRENTICE ENGINEER, BACS DIVISION
Sep 2024 - now

Alter Watt, Paris

- Ran **5** BACS audits solo: compliance with the French BACS decree, the control chain read from sensor to supervision, then recommendations and a cost estimate defended in front of the client.
- Standardised pricing across the division: **5** trade categories, a **3-level** funnel, every price in a single reference tree, mandatory for everyone.
- Wrote a **25-sheet** methodology guide that trains engineers to defend a quote.
- In progress: an offline field-audit web app with quick 3D massing for quantities and one-click report generation.

SHIPWRIGHT INTERN
Jul - Sep 2026

Stirling & Son, Plymouth, UK

- Restoration of a **35-ft** and a **15-ft** wooden vessel: dismantling, diagnosing and hand-making parts to replace damaged structural members.
- Hull preparation, paint and marine-grade protective treatments, two months in a working yard, in English.

APPRENTICE, PV DESIGN
Sep 2022 - Sep 2024

Solar design office

- **100+** PV studies sized and followed, from rooftop to carport. Largest rooftop: **401.4 kWp**, 892 modules on 7 blocks.
- Modelled self-consumption against installed power and landed an extension at **499.95 kWp**, keeping a **120.8 €/MWh** tariff instead of a 40 €/MWh tender cap.
- Designed a frame that did not exist: PV modules held vertically along a parapet next to an airfield. Structural calculation in Robot, site prototype, installation with crews I trained.
- Built the tooling: AutoCAD single-line diagrams, tender files, two VBA tools and a Chrome extension pulling **11** financial KPIs into a scenario comparison.

PROJECTS

SMART PARKING, EDGE IOT
2025 - 2026

ESTP research initiation, team of 4

- ESP32 nodes with ultrasonic sensors and a state machine per space; only confirmed changes go out over MQTT. **< 200 ms** end to end and **≈ −95 %** bandwidth against a 1 Hz stream, in Wokwi simulation; written up with a 26-paper literature review.

BUILDING DESIGN STUDIO
2025 - 2027

ESTP Paris, GECD

- Mixed-use office building, Montrouge: summer comfort without air conditioning, at most **80 h** a year above 28 °C. I lead direction and process and own the KPI workbook; Revit model simulated in Pléiades.
- Bioclimatic student residence: 8,760 hours of climate data, 16 passive strategies, cooling needed only **110 h** a year.

BOLOSS, ONLINE CARD GAME
2026

Personal

- A 1-v-1 browser card game, **71** cards: strict-TypeScript monorepo, server-authoritative state machine on Node and Socket.io, React client.

EDUCATION

ENGINEERING DEGREE, GECD
ESTP Paris · 2024–2027
Apprenticeship track

BUT, THERMAL AND ENERGY ENGINEERING
IUT d'Évry · 2021–2024
Energy audit of a teaching building: 33 kWp PV plant sized, 5-year payback. Compact chip thermal model fitted on 17 FEM runs, ≈ 6 % error.

BACCALAURÉAT, WITH HONOURS
Engineering science · 2021

SKILLS

BUILDING SYSTEMS
BMS / GTB · BACS DECREE · MODBUS · HVAC CONTROLS · ENERGY AUDIT

ENERGY & THERMAL
PLÉIADES · CLIMAWIN · FEM · PID CONTROL · ARCHELIOS · PV*SOL · PVGIS · SOLAREDGE DESIGNER

DESIGN
REVIT · AUTOCAD · SKETCHUP · ROBOT · BIM

CODE
TYPESCRIPT · JAVASCRIPT · VBA · APPS SCRIPT · C++ / ESP32 · MQTT

LANGUAGES

FRENCH
Native

ENGLISH
Professional, UK internship

Every project above, with photos and figures: allonzo-pensabene.netlify.app